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University of Information Technology & Sciences

Lab Report

Department: Department of Computer Science and Engineering

Course Title: Computer Networks Lab

Course Code: CSE0612314

Experiment No.: 04

Experiment Title: Router Configuration using CLI.

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Laboratory Report

1. Experiment Number: 04

2. Experiment Title: Router Configuration using CLI.

3. Objectives:

- To learn how to access the router CLI using Cisco Packet Tracer.
- To understand basic IOS commands used for router configuration.
- To configure a hostname and passwords on the router.
- To assign IP addresses to router interfaces using CLI commands.
- To verify the configuration using show running-config and show ip interface brief.

4. Theoretical Background:

Cisco IOS (Internetwork Operating System) is the software that runs on Cisco routers and switches, and it can be accessed through the Command Line Interface (CLI) to configure and manage the device. Unlike the GUI-based configuration used earlier, CLI configuration involves typing specific commands to set up the router manually, which gives more control and is closer to how real network engineers work. Basic configuration includes setting a hostname to identify the router, applying passwords to secure console and privileged EXEC access, and assigning IP addresses to the router's interfaces so it can communicate with connected networks. Commands like show running-config and show ip interface brief are used to verify that the configuration has been applied correctly. Learning CLI-based configuration is an important step in understanding real-world router management beyond just simulation.

5. Tools and Software used:

- I. Cisco Packet Tracer,
- II. PC-PT,
- III. PT8200 Router,
- IV. Command Line Interface (CLI),
- V. Wires.

6. Network Topology:

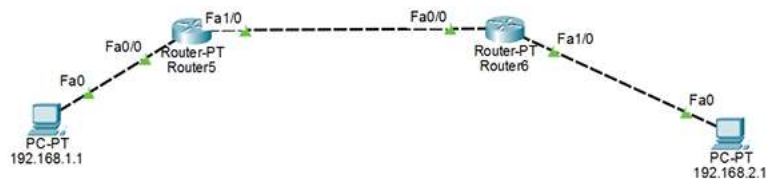


Figure 4.1: PC connected to Router via Console cable for CLI access.

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#exit
Router(config)#int fa1/0
Router(config-if)#ip add 192.168.3.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa1/0
Router(config-if)#ip add 192.168.2.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed
state to up

Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.3.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 4.2: Router1 and Router2 Configuration Using CLI.

```
Router#en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.3.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
        area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C 192.168.1.0/24 is directly connected, FastEthernet0/0
S 192.168.2.0/24 [1/0] via 192.168.3.2
C 192.168.3.0/24 is directly connected, FastEthernet1/0
```

```
Router#en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.3.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
        area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

S 192.168.1.0/24 [1/0] via 192.168.3.1
C 192.168.2.0/24 is directly connected, FastEthernet1/0
C 192.168.3.0/24 is directly connected, FastEthernet0/0
```

Figure 4.3: Router1 and Router2 IP Route.

7. Methodology:

- At first, we opened Cisco Packet Tracer and placed a Router and a PC on the workspace.
- Connected the PC to the router's console port using a console cable.
- Opened the Terminal on the PC through the Desktop tab to access the router CLI.

- Entered privileged EXEC mode using the enable command.
- Entered global configuration mode using the configure terminal command.
- Set the router's hostname using the hostname command.
- Configured console and enable passwords using line console 0 and enable password/secret commands.
- Entered interface configuration mode and assigned an IP address to the router interface using the ip address command, then enabled it with no shutdown.
- Saved the configuration using the copy running-config startup-config command.
- Verified the configuration by running show running-config and show ip interface brief commands.
- Took screenshots of the CLI commands and their outputs for documentation.

8. Output:

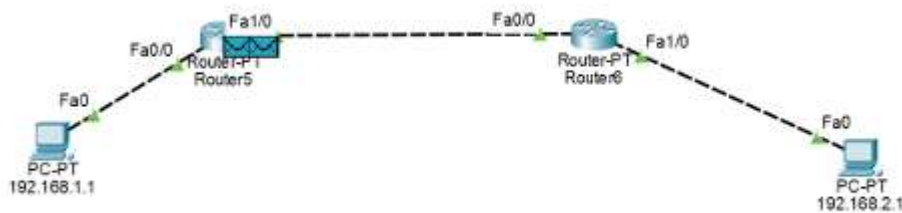


Figure 4.4: Simulation Mode

```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.2.1: bytes=32 time=1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 4.5: CMD Ping Results

9. Conclusion:

This experiment helped in understanding how to configure a router directly through the CLI instead of relying only on GUI-based setup. By setting the hostname, passwords, and interface IP addresses using basic IOS commands, it became clear how routers are actually configured in real networking scenarios. Verifying the configuration with `show running-config` and `show ip interface brief` confirmed that all settings were applied correctly. Overall, the experiment gave hands-on experience with essential CLI commands, which is a fundamental skill for managing Cisco devices.